

Clínica Universidad de Navarra

Principles of Laboratory Medicine

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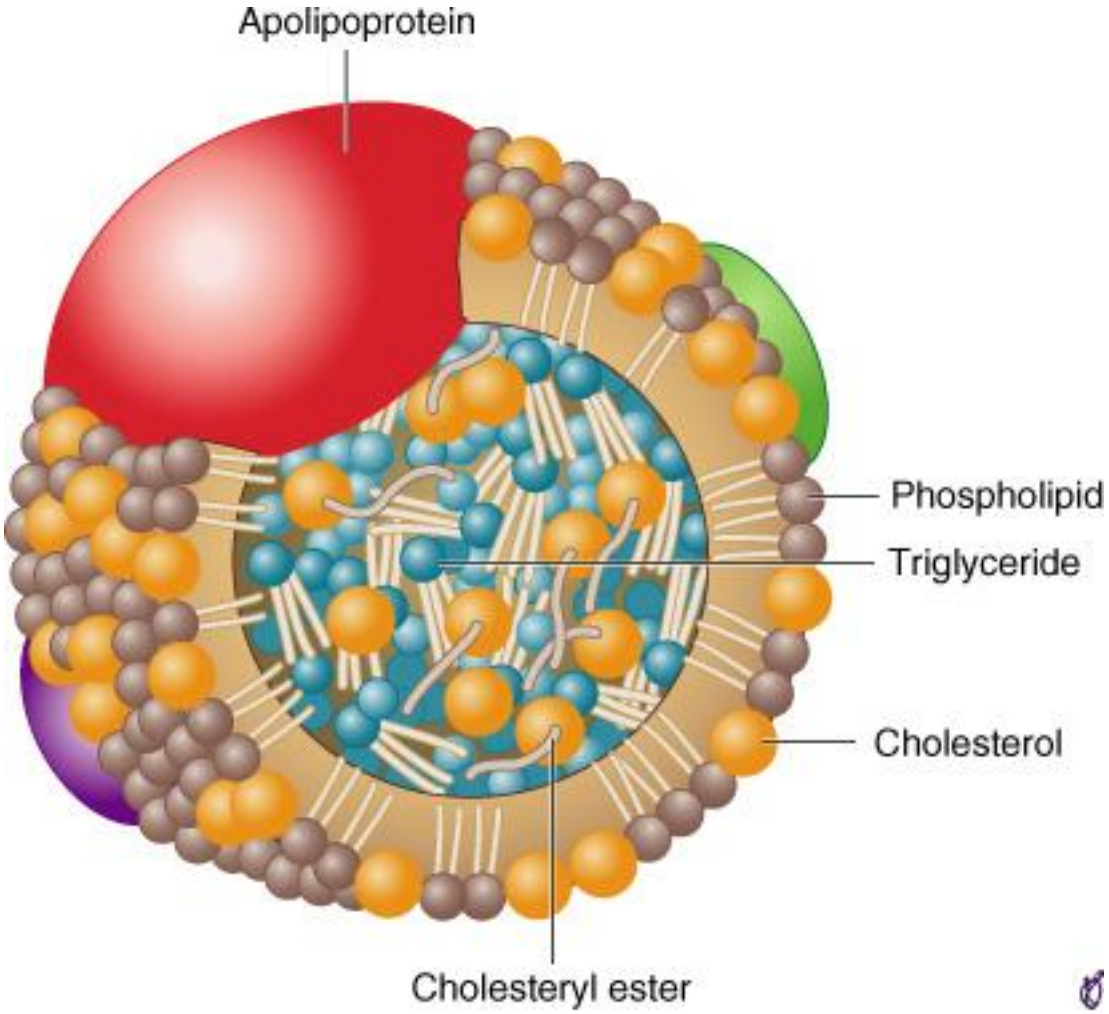
- Major classes of lipoproteins
- Apolipoproteins
- Lipid cycles
- Regulation of cholesterol synthesis

Learning goals

1. Laboratory study of lipid metabolism
2. Lipid metabolism disorders
3. Metabolic Syndrome
4. Cardiovascular risk and atherosclerosis
5. Treatment and monitoring

1. Laboratory analyses to study lipid metabolism

Lipoproteínas plasmáticas



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Lipoproteins functions

Table 67.2 **The four main lipoproteins and their functions**

Lipoprotein	Main apolipoproteins	Function
Chylomicrons	B_{48}, A-I, C-II, E	Largest lipoprotein. Synthesised by gut after a meal. Not present in normal fasting plasma. Main carrier of dietary triglyceride.
Very-low-density lipoprotein (VLDL)	B_{100}, C-II, E	Synthesised in the liver. Main carrier of endogenously-produced triglyceride
Low-density lipoprotein (LDL)	B_{100}	Generated from VLDL in the circulation. Main carrier of cholesterol
High-density lipoprotein (HDL)	A-I, A-II	Smallest lipoprotein. Protective function. Takes cholesterol from extrahepatic tissues to the liver for excretion (reverse cholesterol transport)

Murphy's Clinical Biochemistry 6th ed.

Preanalytical factors

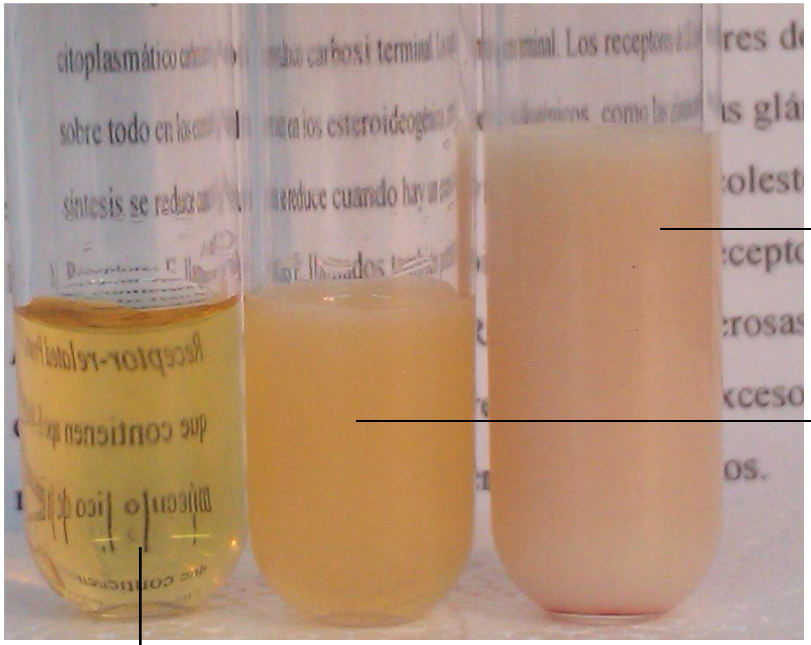
- Diet (↑TG, ↔ COL)
- Exercise (↑HDL-colesterol, ↓LDL-colesterol, ↓TG)
- Alcohol (moderate: ↑HDL-cholesterol; high: ↑TG)
- Pregnancy and lactation
- Diseases: Diabetes mellitus, hypothyroidism, kidney dysfunction, infections...
- Drugs: antihypertensives, immunosupressive agents, sexual steroids…

Recommendations for the patient

In general, the patient has to be "metabolically stable"

- Heep habits at least 2-3 weeks before blood sampling
- Fasting may not be required in the initial evaluation (12 h). Only triglicerides change with recent food intake
- No extreme exercise before blood sampling
- After acute inflammatory diseases wait at least for 2-4 weeks
- Sit for 15 min prior to venipuncture

Visual inspection of the sample



High triglycerides

at 4°C milky layer
↑ Quylomicrons

Normal triglycerides

Who and when to test

Table 1A Lipid determination for assessing cardiovascular risk.[36]

Patients not receiving with lipid-lowering drugs.

1) Routine vascular risk assessment is recommended in patients presenting a major cardiovascular risk factor (i.e. familial history of early CVD, familial hypercholesterolemia, or factors such as tobacco use, arterial hypertension, diabetes mellitus, hyperlipemia, chronic renal disease, obesity or comorbidities that increase the risk for CVD).

2) Consider routine or opportunistic assessment of CVR in men >40 years and women >50 years or postmenopausal women of the general population without CV risk factors.

3) Consider follow-up assessment at 5 years (or earlier if the risk is close to therapeutic thresholds) in all subjects who have undergone screening for CVD risk during an opportunistic screening.

4) Routine CVR assessment is not recommended in men <40 years and women <50 years without known CVR factors.

Monitoring of therapeutic efficacy and adherence to lipid lowering treatment

1) Before lipid-lowering therapy is initiated perform 2 determinations within a 1–2 week interval, except in case of cardiovascular event and in patient with very high risk with indication for immediate treatment.

2) Once lipid-lowering treatment has been initiated, repeat laboratory analysis.

   a. After an acute atherosclerotic vascular event, at 4–6 weeks.

   b. In patients who are stable in cardiovascular terms, at 8 ± 4 weeks until targets are achieved.

3) Once the patient has reached the optimal lipid target, how often should lipids be measured? On a yearly basis

Biochemical tests

1. Basic lipid profile
2. Apolipoproteins
3. Other lipoproteins: Lipoprotein (a)

1. Lipid profile (lipid panel)

	Interpretation	Laboratory
• Triglycerides	→ Atherogenesis	→ Measured
• Total cholesterol	→ Atherogenesis	→ Measured
• HDL-cholesterol	→ Cardioprotection	→ Measured
• LDL-cholesterol	→ Atherogenesis	→ Calculated

LDL estimation: Friedewald equation

LDL-chol = Total chol – HDL-cholesterol – VLDL

LDL-chol = Total chol – HDL-chol – (Triglycerides/5)

Only valid if triglycerides < 400 mg/dL

TG= Triglycerides
Chol = cholesterol

Limitations of the Friedewald equation

1. Not valid if triglycerides > 400 mg/dL

2. If Tg> 200 mg/dL, it is acceptable but calculated LDL underestimates the real LDL

3. Accumulation of methodological errors, as it requires 3 separated analyses

4. It can not be used with very low LDL values

5. Innacurate if Lp(a) is too high

Alternatives to the Friedewald equation

1. Use non-HDL cholesterol as a marker of "atherogenic" cholesterol or apoB

non-HDL cholesterol = total cholesterol– HDL-cholesterol

2. Use Direct methods to measure LDL-cholesterol

3. - If triglycerides > 200 mg/dL report both, estimated LDL and non-HDL colesterol

4. **Martin-Hopkins Equation:**
   - Replaces the fixed number 5 used in the Friedewald estimation (VLDL-C = TG / 5) with variable divisors that depend on triglyceride (TG) levels and non-HDL cholesterol.
   – Provides greater accuracy than the Friedewald formula, especially when TG > 150 mg/dL, LDL-C < 100 mg/dL, and particularly when LDL-C < 70 mg/dL.
   - https://calc4lab.com/calculators/martin-hopkins-ldl

Non-HDL cholesterol

- Therapeutic target in patients with high triglycerides

- Useful for cardiovascular risk assessment as is more closely associated with CVD than other lipid parameters

Case 1: Normal triglycerides

Total cholesterol: 200 mg/dL

Triglycerides: 80 mg/dL

HDL-cholesterol: 40 mg/dL

Clear sample

Calculate LDL-cholesterol

LDL cholesterol (mg/dL) =200 - 40 - 80/5 = 144 mg/dL

Case 2: Triglycerides= 200-400 mg/dL

Total cholesterol: 200 mg/dL

Triglycerides: 320 mg/dL

HDL-cholesterol: 50 mg/dL

Calculate LDL-cholesterol

LDL cholesterol (mg/dL) =200 - 50 - 320/5 = 86 mg/dL

LDL colesterol is underestimated due to the high triglycerides

Also report non-HDL chol=200-50= 150 mg/dL

Case 3. Triglycerides > 400 mg/dL

Total cholesterol: 240 mg/dL

Triglycerides: 420 mg/dL

HDL-cholesterol: 60 mg/dL

Calculate LDL-cholesterol

LDL cholesterol can not be calculated by Friedwald equation

Non-HDL colesterol = 240 – 60 = 180 mg/dL

Use a direct method

Lipoprotein ratios

- Total Cholesterol/HDL (N<4.5)
- LDL/HDL (N<3)
- Higher predictive value than isolated lipoproteins

2. Apolipoprotein measurement

Table 67.1 **Properties of some human apolipoproteins**

Apolipoprotein	Molecular weight	Site of synthesis	Function
A-I	28 000	Intestine, liver	Activates LCAT
A-II	17 000	Intestine, liver	–
B_{100}	549 000	Liver	Triglyceride and cholesterol transport. Binds to LDL receptor
B_{48}	264 000	Intestine	Triglyceride transport
C-I	6600	Liver	Activates LCAT
C-II	8850	Liver	Activates LPL
C-III	8800	Liver	? Inhibits LPL
E	34 000	Liver, intestine, macrophage	Binds to LDL receptor and probably also to another specific liver receptor

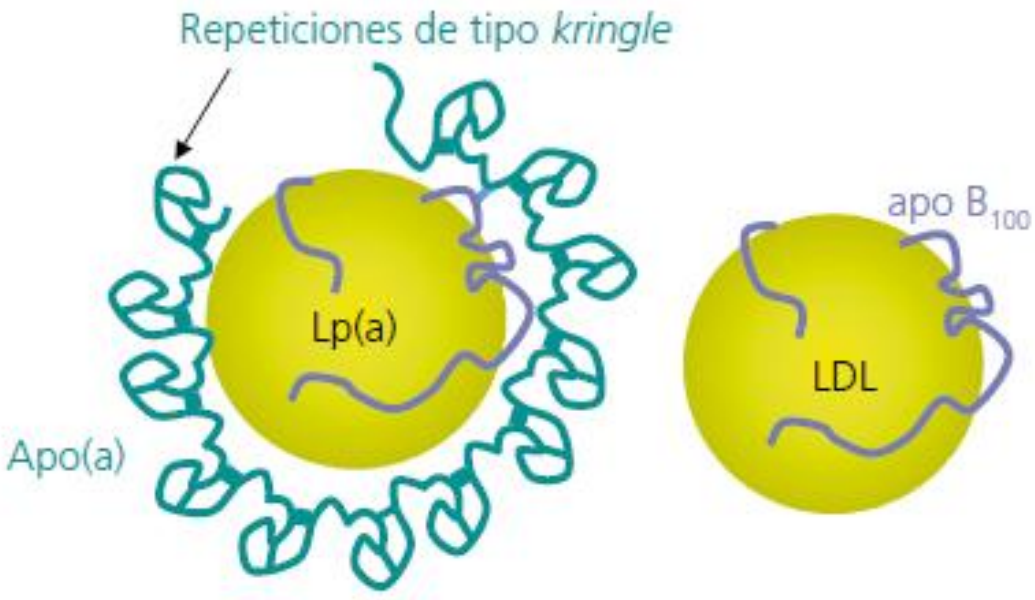
LCAT = Lecithin–cholesterol acyltransferase.
LPL = Lipoprotein lipase.

- apo AI: alternative for HDL
- apo B100: alternative for LDL

Murphy's Clinical Biochemistry 6th ed.

3. Other lipoproteins: Lipoprotein (a)

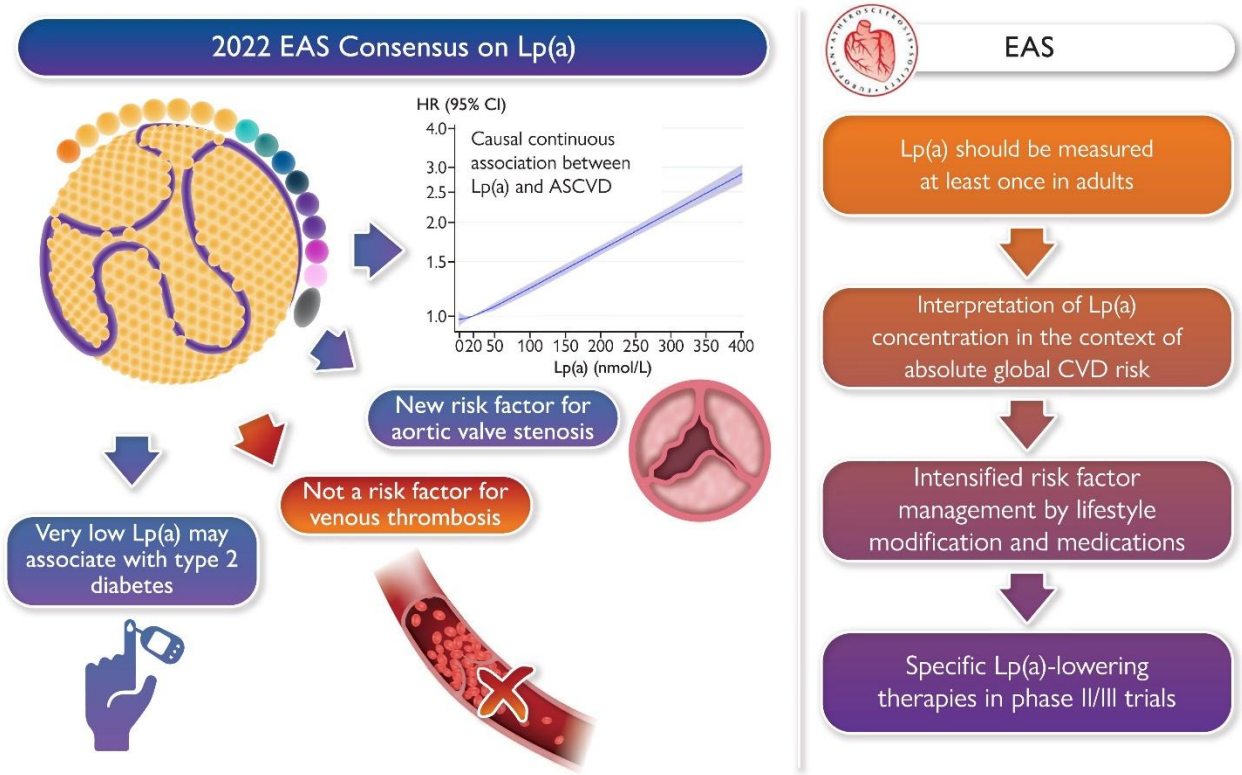
- Contains apo B100. Similar structure to LDL.
- apo (a), similar to plasminogen



Independent cardiovascular risk factor

Increased in patients with kidney dysfuncion

Graphical abstract Key points from the 2022 Lp(a) consensus statement. Current evidence demonstrates a causal continuous ...



Eur Heart J, Volume 43, Issue 39, 14 October 2022, Pages 3925–3946, https://doi.org/10.1093/eurheartj/ehac361

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Laboratory report should include

1. Analytical methods employed and units and when units change
2. Do not use population reference values. Report limits related to the patient´s cardiovascular risk.
3. The use of asterisks in values outside the population reference interval is discouraged.
4. Warning alerts are recommended for extreme lipid levels suggestive of severe dyslipidemia
5. Specific values should be established for the pediatric population.

Arrobas y col. Documento de consenso para la determinación e informe del perfil lipídico en laboratorios clínicos españoles: ¿Qué parámetros debe incluir un perfil lipídico básico? Adv Lab Med 2023; 4(2): 147–156

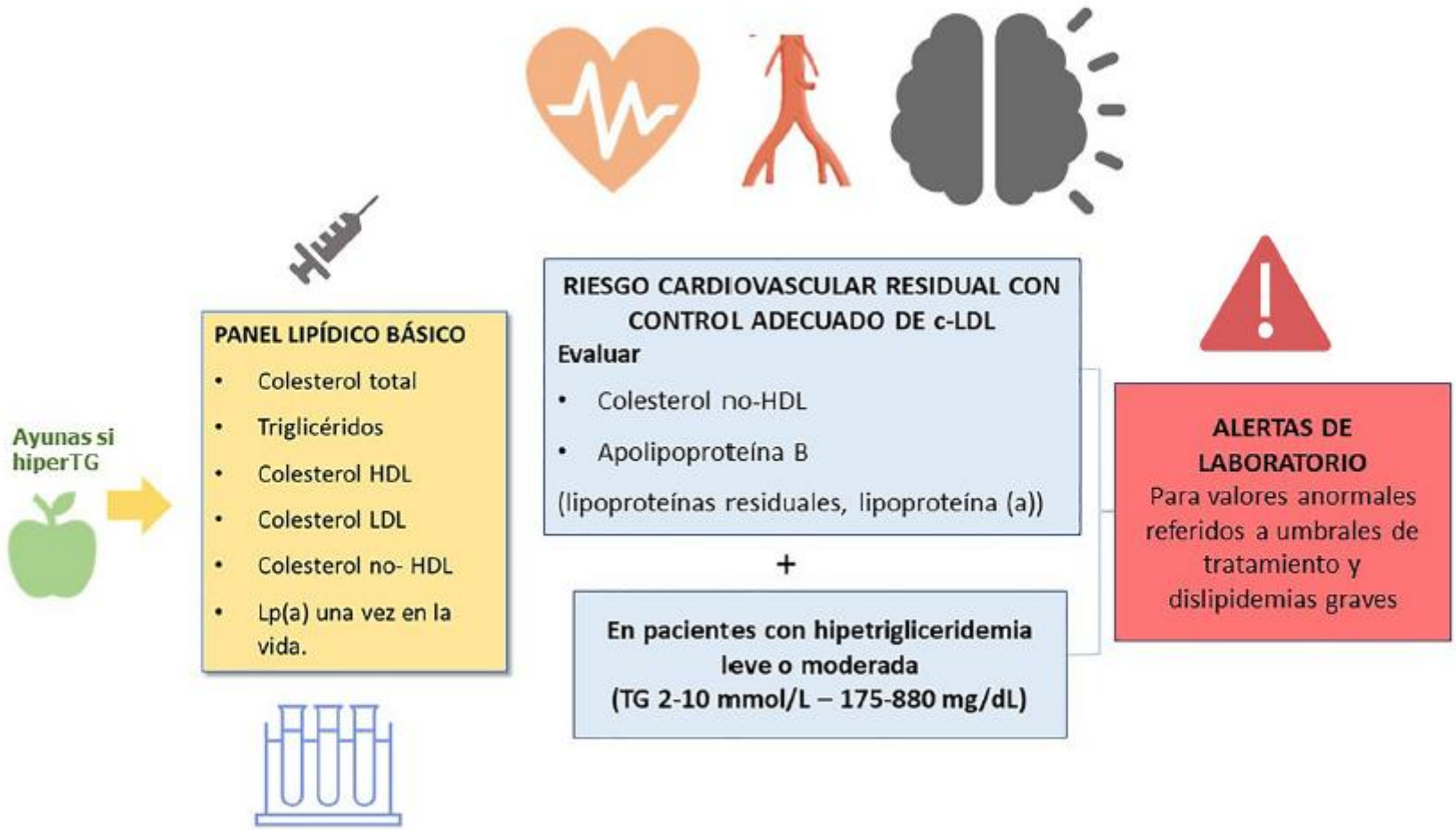


Figura 1: Recomendaciones básicas para el informe de perfil lipídico en laboratorios clínicos españoles.

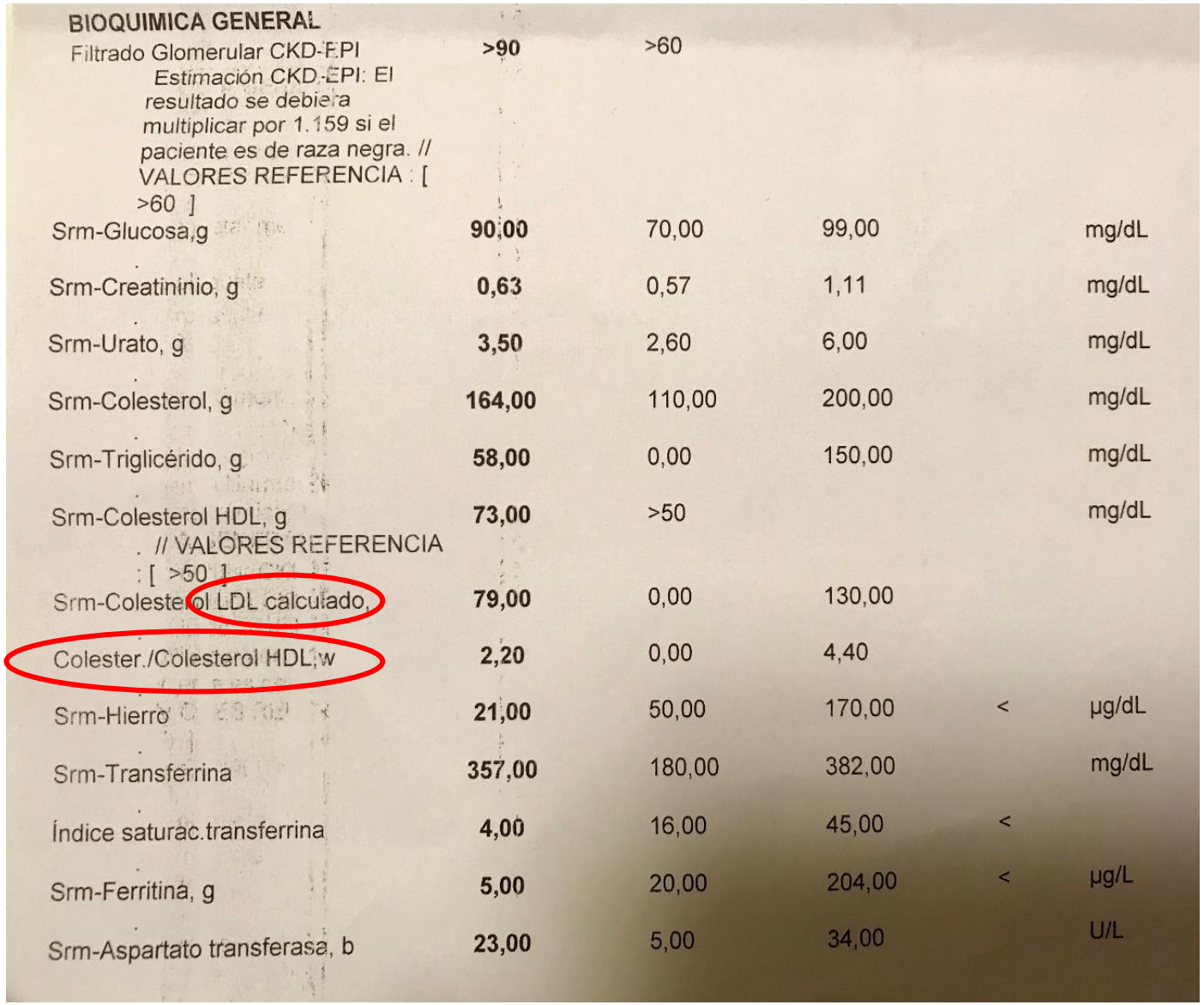
Documento de consenso para la determinación e informe del perfil lipídico en laboratorios clínicos españoles: ¿Qué parámetros debe incluir un perfil lipídico básico?

Tabla 3: Alertas recomendadas para el Sistema informático/Informe de Laboratorio.

Parámetro	Valor crítico	Alerta
Colesterol total	310 mg/dL	Paciente de alto riesgo vascular
Triglicéridos	TG>880 mg/dL	Hipertrigliceridemia severa con riesgo de pancreatitis aguda
Colesterol LDL adultos	>190 mg/dL	Considerar hipercolesterolemia familiar heterocigota
Colesterol LDL adultos	>500 mg/dL	Considerar hipercolesterolemia familiar homocigota.
Triada lipídica aterogénica	Si: TG>150 mg/dL y c-HDL<30 mg/dL c-LDL/ApoB<1,3 o Tg/c-HDL>2	Triada lipídica orientativa de dislipidemia aterogénica de muy alto riesgo vascular
Lipoproteína (a)	>120 mg/dL (260 nmol/L)[a]	Riesgo muy elevado de enfermedad cardiovascular aterosclerosa y estenosis de la válvula aórtica
Apolipoproteína A 1	<10 mg/dL	Valorar Hipoalfalipoproteinemia
Apolipoproteína B	<10 mg/dL	Valorar Hypobetalipoproteinemia

[a]Estimacion según document consenso EAS /EFLM.

Laboratory report



BIOQUIMICA GENERAL					
Filtrado Glomerular CKD-EPI *Estimación CKD-EPI: El resultado se debiera multiplicar por 1.159 si el paciente es de raza negra. //* VALORES REFERENCIA : [>60]	**>90**	>60			
Srm-Glucosa,g	**90,00**	70,00	99,00		mg/dL
Srm-Creatinínio, g	**0,63**	0,57	1,11		mg/dL
Srm-Urato, g	**3,50**	2,60	6,00		mg/dL
Srm-Colesterol, g	**164,00**	110,00	200,00		mg/dL
Srm-Triglicérido, g	**58,00**	0,00	150,00		mg/dL
Srm-Colesterol HDL, g . // VALORES REFERENCIA : [>50]	**73,00**	>50			mg/dL
Srm-Colesterol LDL calculado,	**79,00**	0,00	130,00		
Colester./Colesterol HDL,w	**2,20**	0,00	4,40		
Srm-Hierro	**21,00**	50,00	170,00	<	µg/dL
Srm-Transferrina	**357,00**	180,00	382,00		mg/dL
Índice saturac.transferrina	**4,00**	16,00	45,00	<	
Srm-Ferritina, g	**5,00**	20,00	204,00	<	µg/L
Srm-Aspartato transferasa, b	**23,00**	5,00	34,00		U/L

Laboratory report

Pla-Triglicéridos*	76	mg/dL	≤150
Pla-Colesterol total*	222	mg/dL	≤200
Pla-Perfil lipídico*			
HDL colesterol	73	mg/dL	≥50
LDL colesterol Friedewald	134	mg/dL	≤116
Colesterol no-HDL	149	mg/dL	≤130
Valores de referencia	Valores de colesterol LDL recomendados en función del riesgo cardiovascular: - Muy elevado: < 55 mg/dL - Elevado: < 70 mg/dL - Moderado: < 100 mg/dL - Bajo: < 116 mg/dL Valores de colesterol no-HDL recomendados en función del riesgo cardiovascular: - Muy elevado: < 85 mg/dL - Elevado: < 100 mg/dL - Moderado: < 130 mg/dL		

Interpretación de resultados

Valores de referencia: para el diagnóstico de las dislipemias

- dependen del tipo de población, de la edad y del sexo.
- Para la población española, hay valores de referencia basados percentiles p5 y p95 por edad y sexo.
- Dislipemias familiares monogénicas comienzan en edad infantil. Imp para diagnóstico precoz disponer de valores para esta edad.
- En hospitalizados, mejor esperar a que se resuelva el motivo del ingreso y realizar el estudio en condiciones fisiológicas.

Valores discriminantes: empleados en la práctica clínica para el cribado de la población de las guías NCEP, NICE, EAS, o riesgo individual: regicor, score

Recommended clinical decision limits

Table 2 Desirable lipid values in adults, according to the European Societies of Arteriosclerosis and Laboratory Medicine.[17,18,36]

Parameter	Desirable value in adults
Serum total cholesterol	<200 mg/dL (5.17 mmol/L)
HDL cholesterol	>50 mg/dL women (>1.29 mmol/L) >40 mg/dL men (1.03 mmol/L)
LDL cholesterol	Recommended values based on CVR • Secondary prevention and very-high CVR < 55 mg/dL (<1.4 mmol/L) • High CVR < 70 mg/dL (<1.8 mmol/L) • Moderate CVR < 100 mg/dL (<2.6 mmol/L) • Low CVR < 116 mg/dL (<3 mmol/L)
Non-HDL cholesterol	Recommended values based on cardiovascular risk • Secondary prevention and very-high CVR < 85 mg/Dl (<2.2 mmol/L) • High CVR < 100 mg/dL; (<2.6 mmol/L) • Moderate CVR < 100 mg/dL (<2.6 mmol/L)
Triglycerides	Fasting TG < 150 mg/dL (<1.69 mmol/L) (Non-fasting TG < 175 mg/dL) (<1.97 mmol/L)
Apolipoprotein B	Recommended values based on CVR • Secondary prevention and very-high CVR < 65 mg/dL • High CVR < 80 mg/dL • Moderate CVR < 100 mg/dL (<3.4 mmol/L)
Lipoprotein (a)	<50 mg/dL (<105 nmol/L)
Remanent particles*	<30 mg/dL in fasting conditions <35 mg/dL in non-fasting conditions

CVR: cardiovascular risk.

Consensus document for lipid profile determination and reporting in Spanish clinical laboratories. What parameters should a basic lipid profile include? Neurology perspectives 2023

Recommended Clinical decision limits

	TG (mg/dL)	Total Cholesterol (mg/dL)	LDL-c (mg/dL)	HDL-c (mg/dL)	Non-HDL (mg/dL)
Desirable	< 150 fasting <200 non fasting	< 200		> 40 women > 50 men	
Secondary prevention+ high CVR			< 55		<85
high CVR			<70		<100
Medium CVR			<100		<130
low CVR			<116		

CVR- cardiovascular risk

RECOMMENDATIONS on the determination of the basic lipid profile in cardiovascular prevention

1. Testing lipid profile is not recommended in a context of non-cardiovascular acute inflammatory process. Lipid profile should be determined during the first 24 h after an acute arteriosclerotic ischemic event.

2. Fasting is not required for lipid profile evaluation as an initial cardiovascular risk assessment. If levels of triglycerides exceed Tg ≥ 4.5 mmol/L(≥ 398 mg/dL), a second determination is recommended in fasting conditions for confirmation of results.

3. Reporting the laboratory technique or a change of units is essential for a correct interpretation of laboratory results.

4. The Martin–Hopkins formula replaces number 5 in the Friedewald's formula (c-VLDL = Tg/5) with divisors that vary according to the levels of Tg and non-HDL cholesterol of the patient. The Martin–Hopkins formula is more accurate than Friedewald's when Tg > 150 mg/dL, LDL cholesterol < 100 mg/dL, and especially when < 70 mg/dL. Direct LDL cholesterol assays should be used todetermine LDL cholesterol if Tg ≥ 400 mg/dL.In patients with significantly elevated Lp(a)concentraions, LDL cholesterol estimation should becorrected using the following formula: c − LDL (mg/dL) − [Lp (a) (mg/dL) × 0, 30]

5. On laboratory reports, the reference values for lipid parameters should always be referred to the cardiovascular risk of the patient, rather than to normal reference values for the general population. The use of asterisks marking values outside the population normality range is discouraged. The use of flagging systems is recommended for extreme lipid concentrations suggestive of severe dyslipidaemias. Specific values should be established for the pediatric population

6. The basic lipid profile should include the determination of total cholesterol, HDL cholesterol,triglycerides, non-HDL cholesterol, and LDL colesterol estimation. In patients with mild/moderate hypertriglyceridaemia, testing non-HDL cholesterol and Apo B is recommended for assessing residual cardiovascular risk.

7. Testing Apo B is recommended forassessing vascular risk; classify dyslipidaemias and characterizeparticle size. It is also preferable to non-HDL cholesterol testingin patients with mild-to-moderate hypertriglyceridaemia(175–880 mg/dL), diabetes, obesity, metabolic syndrome, orvery low LDL cholesterol (<70 mg/dL).

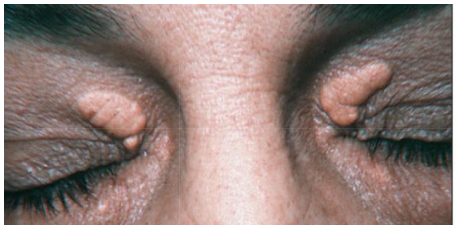
8, Lp(a) should be determined only once in life, except when its levels may be affected by significant changes, as the development of nephrotic syndrome or the use of a therapy to reduce Lp(a).

2. Clinical disorders of lipid metabolism
–Dyslipidaemias–

Clasification of dyslipidaemias

- Hyperlipidaemia o hypolipidaemia
- Elevated lipid: cholesterol, triglycerides, mixed
- Etiology:
  - primary (genetic cause)
  - secondary (lifestyles or medical conditions)

Primary dyslipidaemia



Genetic cause

Polygenic primary dyslipidaemia: more frequent



Types: polygenic hypercholesterolaemia, mixed familial hiperlipidaemia and polygenic hypertriglyceridaemia

Monogenic primary dyslipidaemia: More severe



Very high chol or triglycerides. Family history of early cardiovascular disease. xanthomas, xanthelasma or corneal arcus in < 45 years old.

Types: monogenic hypercholesterolaemia, combined monogenic hyperlipidaemia, monogenic hypertriglyceridaemia, primary hypolipidaemia and primary hypertriglyceridaemia



Secondary dyslipidaemias

- Causes:
  - unhealthy habits
  - underlying disease

- Important to identify them because:
  - Corrected or reduce need for therapy
  - Important health issues or risk factors "*per se*"
  - To confirm or rule out a primary dyslipemia

General strategy for the diagnosis of dyslipidaemias

1. Fasting lipid profile
    a. confirm results 2 or 3 weeks later
    b. Clinical history for cardiovascular risk assessment

2. Evaluate possible secondary dyslipidaemias:
    a. Clinical history, drugs, thyroid function, kidney, liver.
    b. Calculate associated cardiovascular risk

3. If there is no evidence of a secondary cause, *primary dyslipidaemia*

General strategy for the diagnosis of dyslipidemias

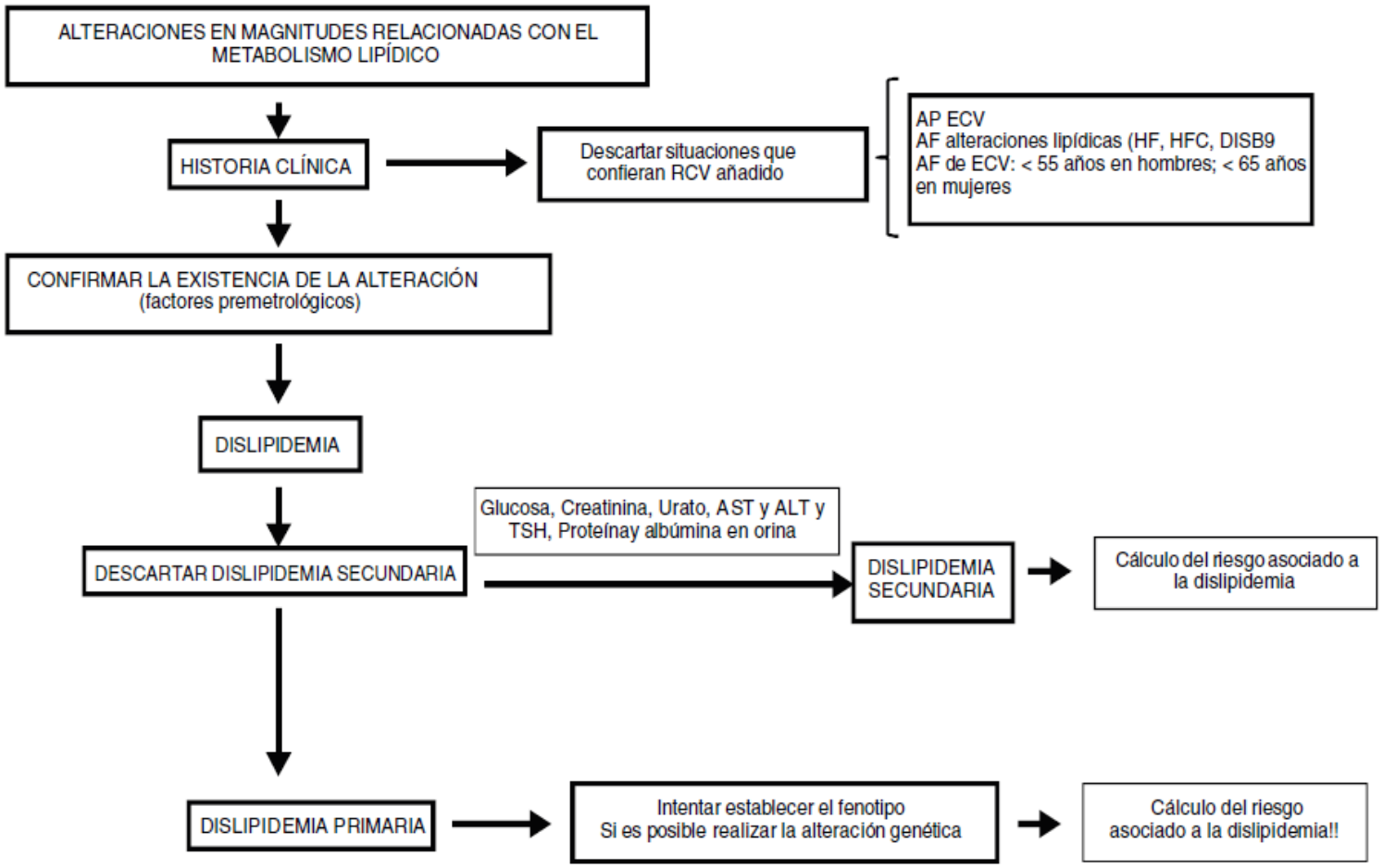


Figura 2 Estrategia general para el diagnóstico de las dislipidemias.

AF: Antecedentes familiares; AP: antecedentes personales; DISB: disbetalipoproteinemia; HF: hipercolesterolemia hipercolesterolemia familiar combinada; RCV: riesgo cardiovascular.

3. Metabolic syndrome

Diagnosis of the Metabolic syndrome at least 3 of the 5 criteria

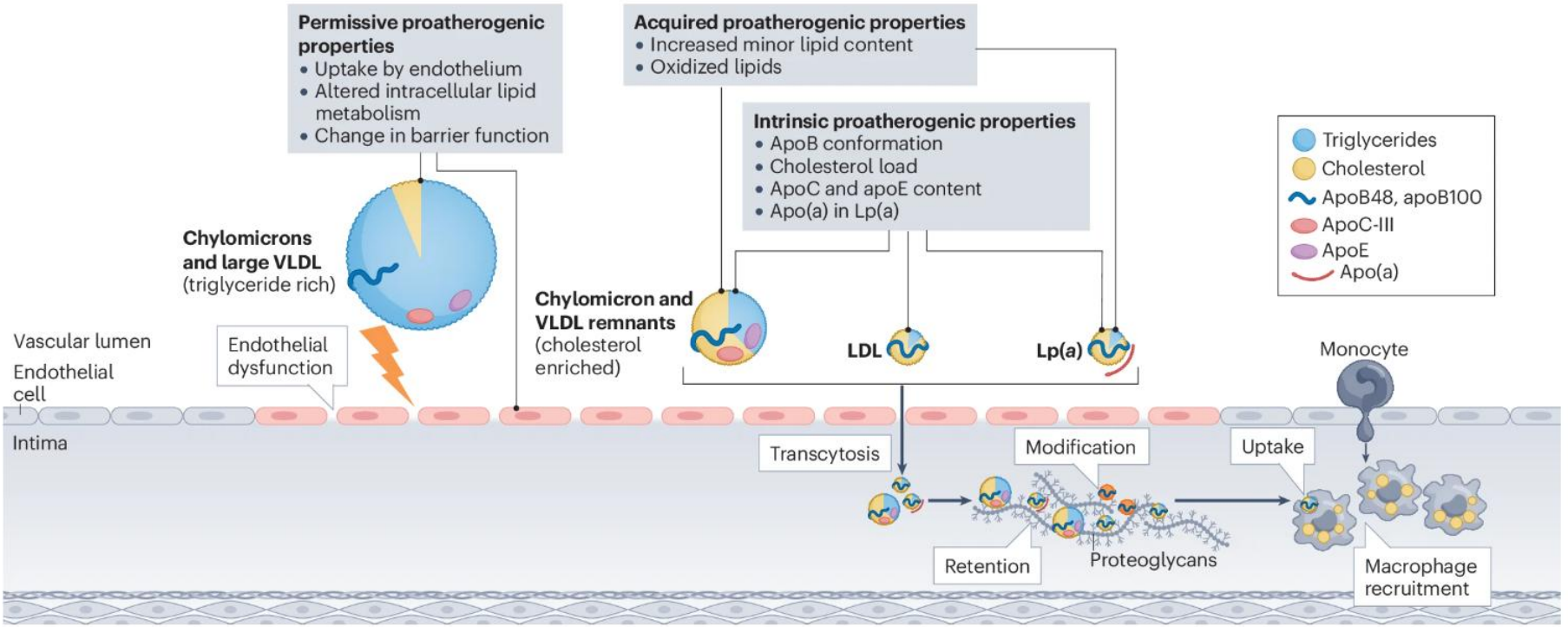
Risk factor	Limit
Abdominal obesity (waist circumference)	
Men Women	>102 cm >88 cm
TG	≥150 mg/dl
HDL-C	
Men women	<40 mg/dl <50 mg/dl
Blood pressure	≥130/≥85 mm Hg
Fasting blood glucose	≥100 mg/dl

Prevalence: 24 % in the general population
Increased risk of diabetes and cardiovascular disease

National Cholesterol Education Program (NCEP) Adult Treatment Panel III (ATP III)

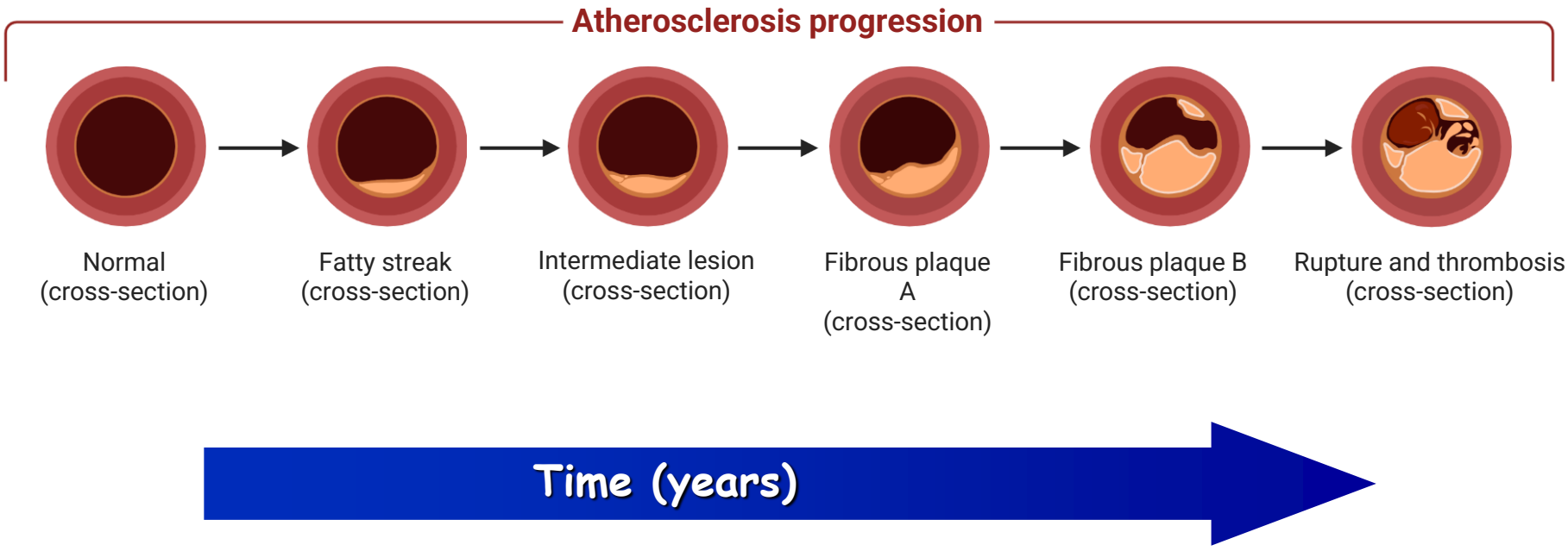
4. Atherogenesis and cardiovascular risk

From: Apolipoprotein B-containing lipoproteins in atherogenesis



Nat Rev Cardiol. 2025 Jan 2. doi: 10.1038/s41569-024-01111-0.
Apolipoprotein B-containing lipoproteins in atherogenesis
Jan Borén 1, Chris J Packardet al

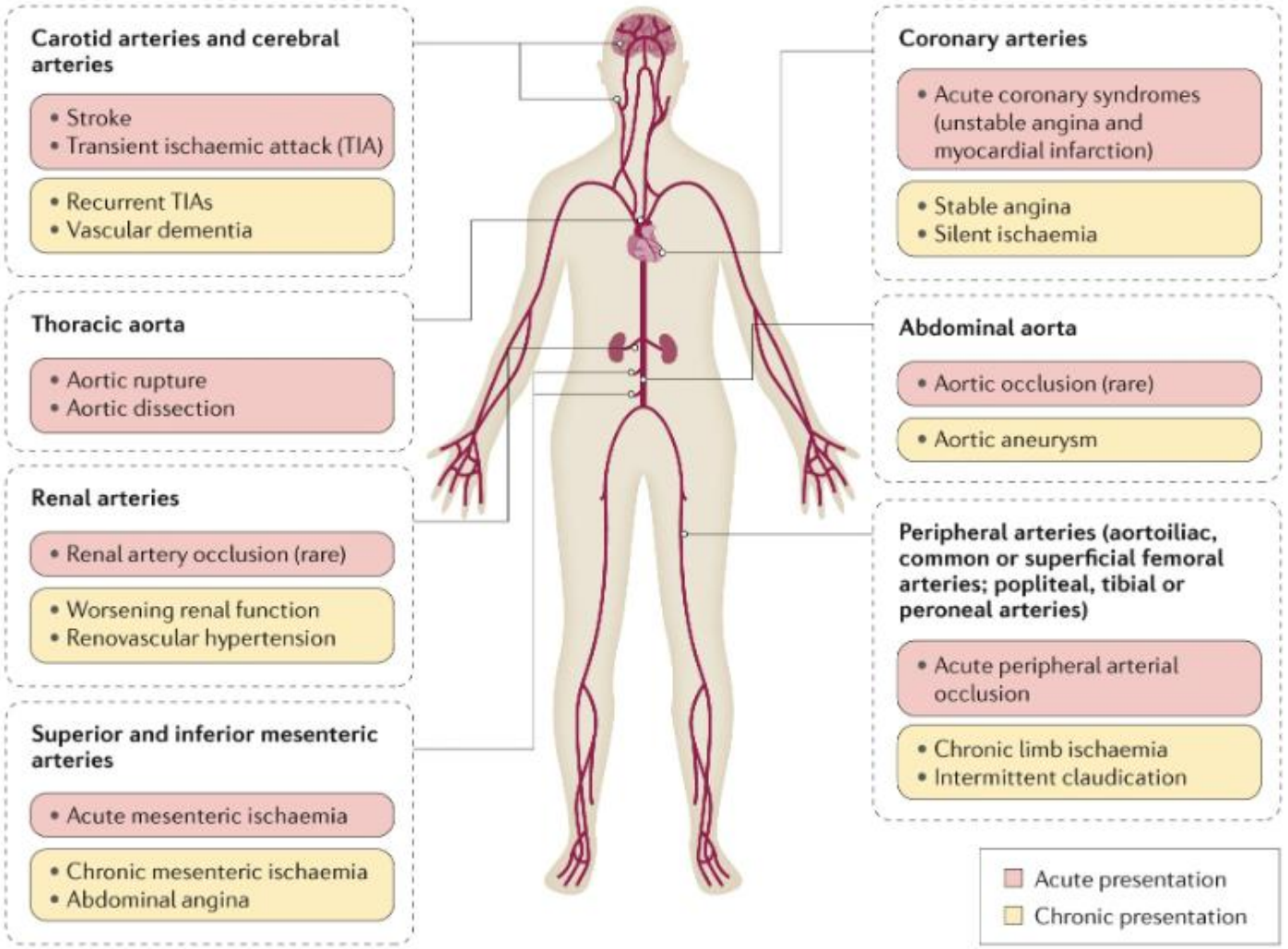
Atherogenesis



Chronic progressive inflammatory process

Fig. 5: Clinical manifestations of atherosclerosis.

From: Atherosclerosis



Atherosclerosis is a systemic disease that may involve multiple vessels. Consequently, the clinical manifestations vary widely according to the vascular territory involved. Despite the systemic nature of many risk factors such as hypercholesterolaemia, hypertension, diabetes mellitus and smoking, atherosclerosis tends to involve primarily specific regions of the arterial tree. Arterial areas subjected to either disturbed flow or low shear stress have particular susceptibility to atheroma formation[41]. These conditions prevail at branch points in the arterial tree.

Cardiovascular risk **Factors**

NON MODIFIABLE	Age Sex Genetic factors
MODIFIABLE	Hypertensión Diabetes mellitus Smoking Alcohol Obesity Physical inactivity High cholesterol

Risk scores

https://heartscore.escardio.org/Calculate/quickcalculator.aspx?model=low

5. Treatment and monitoring

Treatment of dyslipidaemias

- Therapeutic life changes

  Diet
  Exercise

- Drugs

Interpretation of consecutive results

Case

20 years old male with familiar hypercholesterolemia. Starts taking statins:

Visit 1. Total cholesterol= 260 mg/dL

Visit 2. Total cholesterol= 220 mg/dL

Is it a significant change?

RCV for a significant change (95%) = 17.2%

Change: 260-220= 40 mg/dL

% change: (40/260)*100 = 15,3% < 17,2%

Summary

- Lipid profile including total cholesterol, triglycerides, HDL-cholesterol, and LDL-cholesterol is the initial panel in lipid metabolism study.
- Lipid metabolism disorders can be primary or secondary.
- The metabolic syndrome is an accumulation of several disorders, which together raise the risk of an individual.
- Dyslipidaemia is a factor that promotes atherogenesis.
- Treatment of dyslipidaemias could be through therapeutic lifestyle chantes or drugs. The monitoring of the patient should include measuring the change in the lipid profile.